

GAP-10I Torsionful Local Representer: An Explicit Composite-Contortion Right Inverse

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19 July 2026

Abstract

The concurrent-vector obstruction excludes the pure Lorentz pair only in the torsion-free generated-tetrad branch. This note proves an exact complementary result. On every sufficiently small non-null Gaussian patch of a smooth Lorentzian tetrad, one can construct a Lorentz-real biquaternion field Θ and an explicit metric-compatible contortion $K[e, \Theta]$ such that

$$D_\mu \Theta = \sqrt{\mathcal{N}_0} E_\mu, \quad A_\mu = \Omega_\mu(e, K), \quad B_\mu = -\Omega_\mu(e, K)^\dagger.$$

No independent left/right connection fields are introduced. The construction is a local kinematic right inverse of the covariant-tetrad map for an arbitrary smooth metric. It does not prove that canonical UBT dynamics selects this contortion, that the torsion is physically invisible, or that the solution extends globally through horizons or caustics. Accordingly it closes a new local representability subgap while leaving the action-level and global parts of GAP-10I-CURVED open.

1 Geometric setting

Let (U, g) be a smooth four-dimensional Lorentzian patch with a smooth nondegenerate tetrad e_μ^a . Let $\overset{\circ}{\nabla}$ denote the Levi-Civita derivative. Assume that U carries a smooth non-null Gaussian function ρ satisfying

$$g^{\mu\nu} \partial_\mu \rho \partial_\nu \rho = \varepsilon, \quad \varepsilon \in \{+1, -1\}, \quad \rho \neq 0 \text{ on } U. \quad (1)$$

Such a function exists on a sufficiently small Gaussian-normal neighbourhood of any non-null hypersurface; shifting the Gaussian coordinate by a constant ensures $\rho \neq 0$ on a smaller patch.

Define

$$V^\mu := \varepsilon \rho \overset{\circ}{\nabla}^\mu \rho. \quad (2)$$

Then

$$V^2 = \varepsilon \rho^2, \quad V_\mu = \frac{1}{2} \partial_\mu (V^2). \quad (3)$$

2 Composite contortion

Introduce the tensor

$$W_{\mu\nu} := g_{\mu\nu} - \overset{\circ}{\nabla}_\mu V_\nu. \quad (4)$$

Equation (3) gives

$$W_{\mu\nu}V^\nu = V_\mu - \frac{1}{2}\partial_\mu(V^2) = 0. \quad (5)$$

Theorem 2.1 (Explicit metric-compatible contortion). *On the patch $V^2 \neq 0$, define*

$$\boxed{K_{\nu\mu\rho} := \frac{W_{\mu\nu}V_\rho - V_\nu W_{\mu\rho}}{V^2}.} \quad (6)$$

Then $K_{\nu\mu\rho} = -K_{\rho\mu\nu}$, so

$$\Gamma^\nu_{\mu\rho} = \overset{\circ}{\Gamma}^\nu_{\mu\rho} + K^\nu_{\mu\rho} \quad (7)$$

is metric-compatible. Moreover,

$$\boxed{\nabla_\mu^\Gamma V^\nu = \delta_\mu^\nu.} \quad (8)$$

Proof. Antisymmetry of Eq. (6) in the first and third lowered indices is immediate and is precisely the metric-compatibility condition for the difference of two affine connections. Contracting with V^ρ and using Eq. (5) gives

$$K^\nu_{\mu\rho}V^\rho = W_\mu^\nu. \quad (9)$$

Therefore

$$\nabla_\mu^\Gamma V^\nu = \overset{\circ}{\nabla}_\mu V^\nu + W_\mu^\nu = \delta_\mu^\nu. \quad (10)$$

□

The torsion is the composite tensor

$$T^\rho_{\mu\nu} = 2K^\rho_{[\mu\nu]}. \quad (11)$$

No independent torsion variable is needed for this local representer.

3 Single- Θ tetrad representer

Let $\{\mathbf{u}_a\}$ be a Lorentz-real biquaternion basis and write

$$E_\mu = e_\mu^a \mathbf{u}_a, \quad V^a = e^a_\nu V^\nu, \quad s := \sqrt{\mathcal{N}_0}. \quad (12)$$

Define the Lorentz-real field

$$\boxed{\Theta := sV^a \mathbf{u}_a.} \quad (13)$$

Let $\omega(e, K)$ be the frame connection corresponding to Eq. (7), and let $\Omega = \rho_*(\omega)$ be its biquaternionic spin representation. Use the already proved Lorentz pairing

$$D_\mu X = \partial_\mu X + \Omega_\mu X + X \Omega_\mu^\dagger, \quad A_\mu = \Omega_\mu, \quad B_\mu = -\Omega_\mu^\dagger. \quad (14)$$

Theorem 3.1 (Local torsionful right inverse of the tetrad map). *The field in Eq. (13) satisfies*

$$\boxed{D_\mu \Theta = sE_\mu.} \quad (15)$$

Consequently the canonical anticommutator readout reproduces the prescribed metric exactly:

$$\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu} \mathbf{1}. \quad (16)$$

Proof. The tetrad postulate for (e, ω, Γ) gives

$$D_\mu V^a = e^a{}_\nu \nabla_\mu^\Gamma V^\nu. \quad (17)$$

Using Eq. (8),

$$D_\mu \Theta = s(D_\mu V^a) \mathbf{u}_a = s e^a{}_\nu \delta_\mu{}^\nu \mathbf{u}_a = s E_\mu. \quad (18)$$

The metric identity then follows from the Lorentz Clifford anticommutator. $\square$

Corollary 3.2 (Local existence for arbitrary smooth Lorentzian tetrads). *Every point of a smooth Lorentzian tetrad manifold has a sufficiently small patch on which a shifted non-null Gaussian coordinate satisfies Eq. (1) with $\rho \neq 0$. Hence every smooth tetrad has a local single- Θ representer of the form above with a composite metric-compatible contortion.*

4 Schwarzschild vacuum exterior

The result is about the metric geometry and does not require interpreting the central object as an exactly nonrotating physical black hole. On any exterior patch $r > 2M$ of the non-flat Schwarzschild vacuum metric, let

$$f(r) = 1 - \frac{2M}{r}, \quad \rho(r) = \rho_0 + \int_{r_0}^r \frac{du}{\sqrt{f(u)}}. \quad (19)$$

Then $g^{-1}(d\rho, d\rho) = +1$ and

$$V = \rho \partial_\rho = \rho \sqrt{f(r)} \partial_r. \quad (20)$$

Equations (6) and (13) therefore give an explicit local torsionful representer of the Schwarzschild vacuum exterior with $M \neq 0$. This does not contradict the torsion-free no-go: the constructed connection has, in general, nonzero composite torsion.

5 What this does and does not close

Exact status after the torsionful construction

GAP-10I-TORSION-LOCAL: CLOSED LOCALLY [L1]. For every smooth Lorentzian tetrad, a local non-null Gaussian patch admits an explicit Lorentz-real single- Θ representer with $A = \Omega(e, K)$, $B = -\Omega(e, K)^\dagger$ and composite metric-compatible contortion (6). No independent A , B , or propagating torsion field is introduced at the kinematic level.

GAP-10I-PAIR-GR: CLOSED AS A TORSION-FREE NO-GO [L1]. The concurrent-vector obstruction remains valid when $K = 0$. It must not be stated as a no-go for the same pure Lorentz pairing with arbitrary composite torsion.

GAP-10I-2S: NOT REQUIRED FOR LOCAL KINEMATIC REPRESENTABILITY. A nontrivial relative left/right pair (P, Q) is one possible torsion-free completion, but it is not the only local curved route. Composite contortion already supplies a local right inverse inside the one-connection pairing.

GAP-10I-CURVED: LOCAL KINEMATICS CLOSED; DYNAMICS/-GLOBAL PART NARROWED. The construction does not show that the canonical UBT action selects Eq. (6), that the induced torsion is compatible with all matter couplings and observations, or that the representer extends through zeros of V^2 , Gaussian caustics, horizons, and global topology.

GAP-10D and GAP-U2 Θ : NOT CLOSED. Einstein dynamics, the exact spin current, action-level algebraic elimination of torsion, physical torsion constraints, and on-shell selection of the Schwarzschild/Kerr/FRW sectors remain separate dynamical problems.

References

- [1] R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- [2] F. W. Hehl, P. von der Heyde, D. G. Kerlick, and J. M. Nester, “General Relativity with Spin and Torsion: Foundations and Prospects,” *Rev. Mod. Phys.* **48**, 393–416 (1976).